
How the Two Models Connect

The economic channels of the OG-Core $\times$ CLEWS soft link —
and how each is represented in the model

numbers: the Philippine reference run · `ogclews-link` · July 6, 2026



The coupling problem

Two models, complementary blind spots

CLEWS / OSeMOSYS

a least-cost optimiser: chooses what to build and run so that energy demand is met at the lowest discounted system cost

- **rich:** technology, season-by-season operation, emissions, land & water
- **blind:** demand is *fixed* — no price response, no households, no government budget, no distribution

Each is detailed exactly where the other is blank — so what usefully crosses the seam differs by direction:

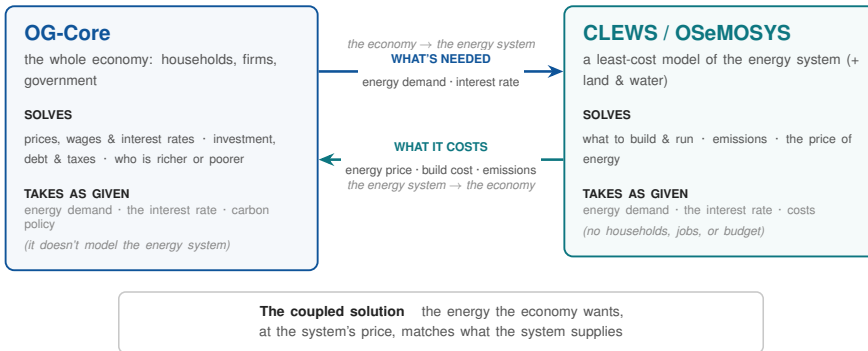
quantities flow from the economy, **prices and physical outcomes** flow from the energy system.

OG-Core

overlapping-generations general equilibrium: households by age $S \times$ lifetime income J , M industries, a government that closes its budget each period

- **rich:** behaviour, saving, fiscal policy, incidence, demography
- **blind:** no energy technology — energy is not a priced input to production

The structural seam



Each model is hollow exactly where the other is deep.

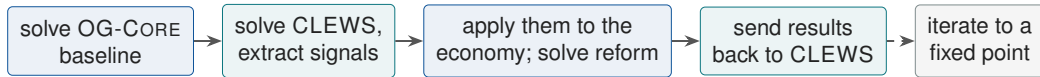
Quantities flow right (energy demand, the interest rate); prices and physical outcomes flow left (the energy price, build cost, emissions). The models stay separate and pass a few auditable signals.

Why a *staged soft* link, not one big model

A hard link is ruled out structurally, not practically:

- OG-CORE steps through time, period by period; the energy model plans all years at once
- neither fits inside the other — OG-CORE has no single objective the energy model could be nested in — and no shock can be injected mid-solve

So the models stay separate and the link runs in stages:



Validated today: the single pass (solid). The loop (dashed) is designed — the return signals already exist — but not yet wired.



Signals, wedges, and the price that carries the load

The vocabulary: signal → wedge → channel

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Wedge the OG-CORE parameter that carries it: the consumption tax τ^c , an industry's productivity Z_m , public investment α_I , mortality $\rho(s, t)$, effective labour $e(t, s, j)$. (A “wedge” in the tax sense — the simplest channel inserts one between producer and consumer price.)

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Channel one signal mapped to one wedge, with a guard that fires when the mapping is large or ill-posed.

Each channel is one small, testable mapping — and nothing country-specific is built in: which industry is electricity, and how many industries there are, is read from the country's own model at run time.

The load-bearing signal: the system's own marginal cost

The energy model must balance supply and demand for every fuel, in every year and season. Ask how much total system cost would rise if one more unit had to be supplied, and you get that fuel's **marginal cost** — under the usual competitive reading, *the price that quantity would fetch in a market*.

1. It supplies the missing price response

Inside the energy model, demand is fixed. Hand the price to OG-CORE households — whose energy use responds to price, down to a subsistence floor — and you get the response the optimiser lacks.

2. It reveals incidence

A single “cost” is silent on who bears it. $S \times J$ households distribute the burden; because energy is a partial necessity, an energy-price change is *regressive* — invisible to a single-agent model.

When are the two models consistent? (the target)

$D^* = D_{\text{OG}}(\pi^*)$ and $\pi^* = \pi_{\text{C}}(D^*)$: the economy demands D^* at price π^* , and π^* is the system's marginal cost of delivering D^* . A single pass uses the price at *baseline* demand; iterating to the fixed point closes the loop.

Three routes for one energy cost

OG-CORE production has no energy input — an energy-cost signal must enter through one of three routes:

- A a tax wedge on household energy** — raise the price households pay by the reform/baseline ratio. Reuses existing structure and shows incidence; but it touches only household energy, and its “revenue” is phantom. Empirically $\approx$ output-neutral.
- B a supply-side cost-push** — raise each industry’s cost by the energy it uses, directly and through what it buys from others, as a productivity cut. Right sign and materially contractionary; but firms cannot substitute away, so the size is only a proxy.
- C energy as a priced input to production** — firms buy energy alongside capital and labour, so the price feeds their costs directly and the loop closes on quantities too. The structural ideal — needs an extension of OG-CORE (unbuilt).

The route is a first-order modelling choice: routes A and B carry the *same* physical cost to opposite macro verdicts — roughly neutral vs. sharply contractionary. The structurally faithful route (C) is therefore the target, not a refinement.



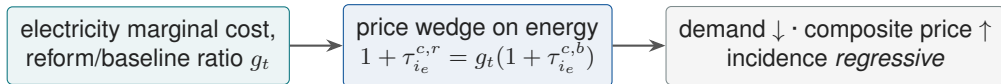
The eight channels

The map

Channel	Dir.	CLEWS signal	OG-CORE object	First-order effect
Energy price	c→o	electricity marginal cost (reform/baseline)	price wedge (τ^c) on the energy good	demand ↓; regressive incidence
Carbon price	pol	carbon price (\$/tCO ₂)	energy-wedge add-on + emissions penalty	demand ↓; emissions ↓
Public investment	c→o	grid & transmission build cost	public-investment share → public capital	public capital ↑; crowding-out, debt
Capital intensity	c→o	power-fleet capital-cost share	energy industry's capital share (γ_m)	energy price ↓; energy K ↓
Investment incentive	pol	tax-credit rate (policy response)	cost of capital (investment tax credit)	capital reallocates into energy
Health	c→o	PM _{2.5} emissions change	survival rates + worker productivity	mortality ≈ 0 ; morbidity ↑ labour
Discount rate	o→c	—	economy's return → planning discount rate	shifts the build mix (via the loop)
Energy demand	o→c	—	economy's activity → energy demand	more capacity (via the loop)

four CLEWS→OG-CORE · two policy levers applied once to both sides · two OG-CORE→CLEWS

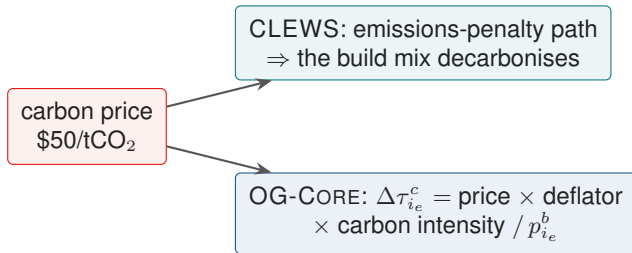
Energy price — the system's marginal cost reaches household budgets



- households face $(1 + \tau_{i_e}^c) p_{i_e}$, so energy demand responds — the price response CLEWS lacks
- poorer households spend more of their budget on energy, so they bear more of the change — even when the revenue is returned (recycled variant: $C_{ss} = -0.087\%$)
- the price is the scenario's *real* electricity price, which for the Philippines is **near-flat** — so the standalone effect is tiny ($Y_{ss} = -0.0005\%$): transmitted honestly, the data does little
- the earlier $\pm 20\%$ runs only stress-tested the *transmission*

maturity: route A, on the real price; the coupled run transmits the same price through the composite (later).

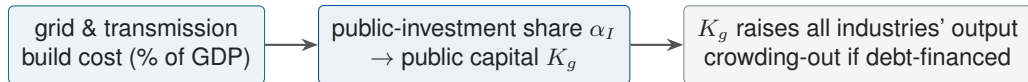
Carbon price — one price, two sides, counted once



- one chosen price enters *both* models along the same path — never read back a second time from the energy model's own implied carbon cost
- OG-CORE firms buy no energy, so only *household* energy carbon is priced on the macro side; industrial carbon is priced in CLEWS
- revenue is recyclable (transfers); standalone it is the clearest contraction: $Y_{ss} = -0.040\%$

the *level* rides on the uncalibrated unit/deflator bridge — mechanics validated, magnitude illustrative.

Public investment — the grid on the government's books



- this is where crowding-out *legitimately* lives: debt-financed spending competes for the savings pool, raises r , displaces private investment
- public-grid capital and private-generation capital touch different objects — coupling both is complementary, not double counting
- in the Philippine scenario the *extra* grid spending is ≈ 0 (the transition is generation-side), so the channel is inert here ($Y_{ss} = +0.000\%$); the mechanics scale with any non-zero spending

The capital pair I: capital intensity — a worked example

Signal: reform/baseline ratio of the power fleet's capital-cost share $\Rightarrow$ scales the energy industry's capital exponent γ_m (here +12.2%).

The model's own first-order condition (Cobb–Douglas, $\varepsilon = 1$): the user cost ρ_m and taxes cancel in the reform/baseline ratio, leaving an exact identity —

$$K_m = \frac{\gamma_m p_m Y_m}{\rho_m} \quad \Rightarrow \quad \hat{K}_E = \hat{\gamma}_E \cdot \hat{p}_E \cdot \hat{Y}_E$$

	$\hat{\gamma}_E$	$\hat{p}_E$	$\hat{Y}_E$	$\hat{K}_E$ (obs.)	identity
energy industry ($m_e = 2$)	+12.2%	−42.8%	+20.9%	−22.4%	−22.4%

- the higher exponent cuts the sector's unit cost, and competition passes that into price $\rightarrow$ electricity **collapses** −42.8%
- cheaper power lifts output +20.9% — but revenue $p_m Y_m$ still shrinks, so capital **falls** −22.4% (the identity closes to 0.03%)

Raising the capital *share* sheds capital: this is a factor-share and energy-price lever, *not* a capital draw-in.

The capital pair II: the investment incentive — the actual draw-in lever

The capital-*attraction* force is a structurally different instrument: an **investment tax credit** on the energy industry.

$$\rho_m = \frac{r + \delta - \tau_m^b \delta^\tau - \tau_m^{\text{inv}} \delta}{1 - \tau_m^b}$$

the ITC enters the *user-cost denominator*, where γ_m is absent $\Rightarrow$ capital demand shifts *out* at given r

the capital share γ_m (capital intensity)

moves factor *demand* through the production exponent; absent from ρ_m . Result: cheaper electricity, *less* energy capital.

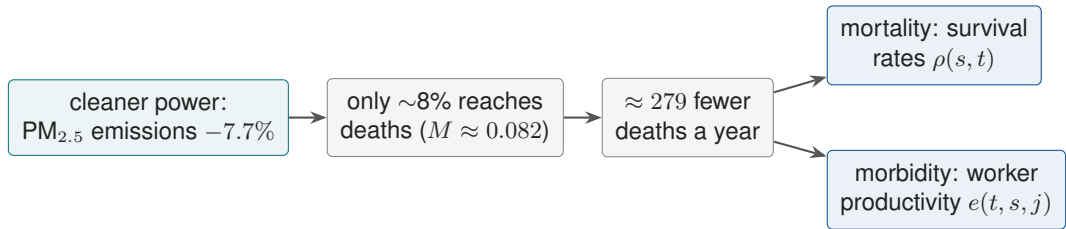
the tax credit (investment incentive)

moves the *cost of capital*; absent from the exponents. Result: capital reallocates *into* energy. (A defined lever; not part of the Philippine reference run.)

The lesson

Two levers, two different objects — *not* two views of one build-out. Intuition says “capital-intensive $\Rightarrow$ pulls capital in”; the model’s own first-order condition shows the opposite. Don’t stack them.

Health — cleaner air, fewer deaths, more productive workers



- real inputs: the Global Burden of Disease counts **43,951** deaths a year in the Philippines from breathing fine particulates (PM_{2.5}), with real age profiles
- cutting power-sector PM_{2.5} by 7.7% is *not* 7.7% fewer deaths: energy is one source among many, and deaths fall less than one-for-one as the air clears — so only ~**8%** of the cut reaches mortality ($M \approx 0.082$). That is **≈279 fewer deaths**, not the ≈3,400 from blaming energy for *every* pollution death
- **mortality**: pollution deaths fall mostly among the elderly, so the lives saved are mostly retirees — long-run output barely moves
- **morbidity**: fewer working-age illnesses means more productive workers — and *that* is the real macro gain (+0.19% near-term; steady state -0.003%)

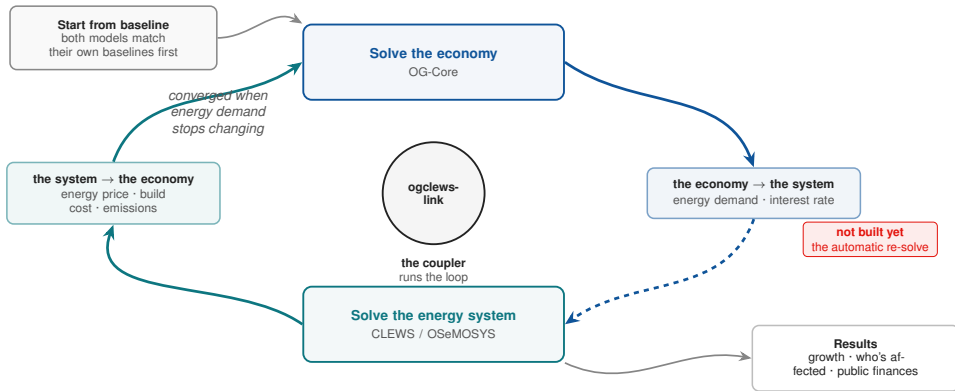
What the economy sends back: its return, its demand

Two channels run *after* the economy solves and write ready-to-use CLEWS inputs:

- **discount rate:** the economy's equilibrium return r_p becomes the energy system's planning discount rate — so energy planning uses the *model-consistent* cost of capital. A lower return favours capital-heavy, long-lived assets.
- **energy demand:** how activity changed (industry output or consumption, with an elasticity) scales the demand the system must meet. In a single pass the change is $\approx$ zero — latent by construction.

They move the macro economy only once the loop iterates — the quantity link of the classic coupled energy–macro models (MESSAGE-MACRO, TIMES-MACRO), but feeding an economy with households and a government instead of a single representative agent.

Closing the loop



Re-solve the two models in turn until energy demand stops changing. Validated today: the single pass; naive price feedback can oscillate, so the loop needs damping.



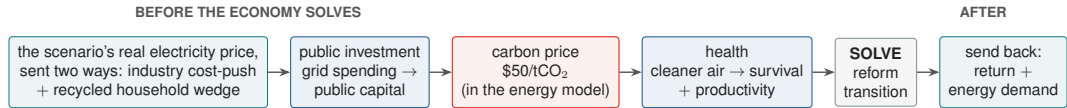
Counting each signal once

Five principles, each enforced by a guard

1. **One carbon price.** It enters once — a CLEWS penalty and an OG-CORE wedge on the same path — never *also* read back from the energy model's implied carbon cost.
2. **One view of an energy-cost shock.** A price wedge, a carbon wedge, and a productivity haircut are alternative doors for the *same* cost — never stacked.
3. **One view of generation capital.** A cost-push *or* an investment credit — and the capital share (γ_m) is *not* a third interchangeable view: it is a factor-income lever that sheds capital (the worked example).
4. **Revenue must be routed.** A τ^c wedge mechanically raises consumption-tax revenue; recycle it (transfers, public investment) or finance it explicitly — else a hidden fiscal expansion rides along.
5. **Public and private capital are distinct.** $\alpha_I \rightarrow K_g$ and the firm's own K touch different objects; coupling both is complementary.

Does it behave? The Philippine coupled run

Anatomy of the flagship: the coupled run

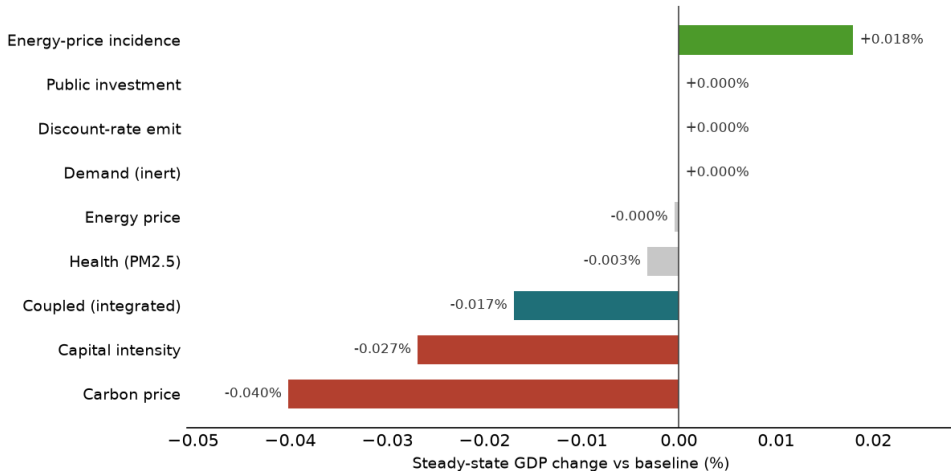


- the Philippine electricity price is near-flat $\Rightarrow$ the energy channels contribute almost nothing (electricity output moves -0.18%); **the macro response is carried by health**
- here the carbon price acts in the energy model (one-price principle); the -0.040% shown earlier is the standalone household-wedge version

% change from baseline	steady state	t_0	t_{10}
output Y	-0.017	$+0.165$	$+0.395$
consumption C	$+0.008$	$+0.195$	$+0.467$

PHL OG×CLEWS — steady-state macro impact by channel

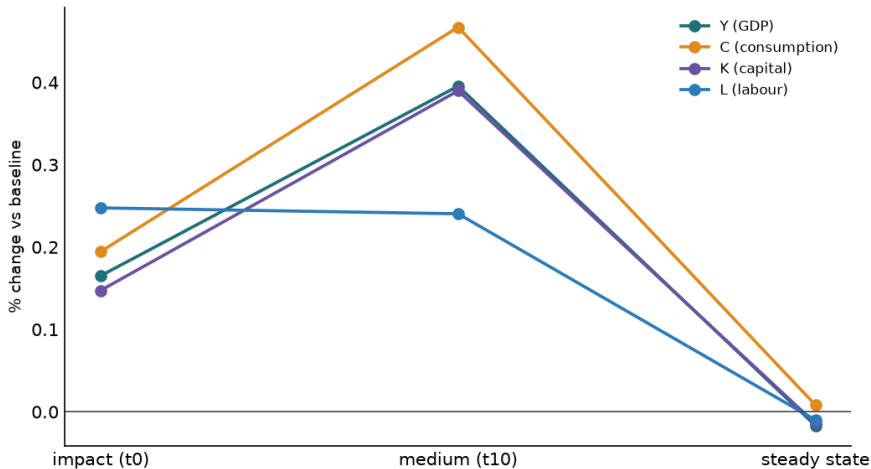
M=8 · ogcore 0.16.3 · real CLEWS 'auto' electricity price (no illustrative +20%)



Per-channel steady-state output impact, each channel run alone on the Philippine calibration. The energy price is ≈ 0 on the near-flat real price; the carbon price (-0.040%) is the clearest contraction; capital intensity is -0.027% in aggregate (but reshapes the energy sector); health is -0.003% at steady state yet $+0.19\%$ near-term.

PHL coupled result over the transition

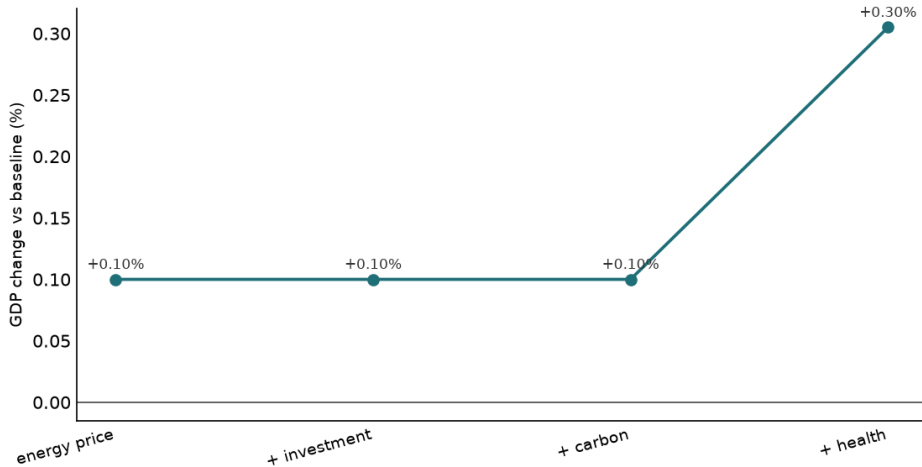
real CLEWS energy + public investment + carbon + GBD health · M=8 · ogcore 0.16.3



The coupled transition: steady state -0.017% , but $+0.165\%$ at t_0 rising to $+0.395\%$ by t_{10} — the working-age morbidity gain along the path, added retirees in the long run. The horizon split is the health channel's signature, and it is invisible to a static model.

Cumulative channel contributions to the real coupled result

each step adds one channel; top step = coupled · M=8 · real CLEWS energy



Cumulative decomposition: channels layered so the top layer *equals* the coupled run — every effect attributable to a named mechanism. That auditability is the point of the channel architecture.

Status

Where the coupling stands

Validated today

- eight channels, each one signal → one wedge, guard-railed
- single CLEWS→OG-CORE pass on *real* inputs: the scenario's real electricity price, the real Global Burden of Disease data
- each model runs in its own software environment; the link passes a few auditable files between them
- every number locked in a reference record and re-checked by tests; signs and mechanics as theory predicts

Open, in order

- **loop closure** — the emitters are ready; the outer re-solve-and-iterate is not wired
- **the unit/deflator bridge** — carbon and investment *levels* are illustrative until it is calibrated
- **Route C** — energy as a priced production input: the structural destination

One line: A small, auditable library of economic mechanisms — each signal counted once, each effect attributable — rather than one opaque coupling.